

Project submission: what to hand in and how

How to deliver the big project (distributed MapReduce system). Read this carefully **before the last session**. One submission per group.

Where to submit

Everything goes into a **single Git repository per group**:

- Create a Git repository (e.g. on GitHub).
 - Push **all** the deliverables listed below into it.
 - **Give me access** to the repository (add me as a member) **or** send me the repository URL — make sure I can actually open it.
 - **Deadline: push the final state and give me access at the latest one week after the last session.**
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What to submit

1. The written report

The report must cover:

- the **architecture** of your system,
- the **protocol design** (with your space-time diagrams),
- the **performance metrics**,
- the **Amdahl's law** graph,
- the **pain points**: what went wrong and why,
- the **3 use cases** (beyond word frequency) and the explanation of their results,
- the **comparison** of your system vs Hadoop vs Kafka Streams (batch vs stream processing).

Format: any readable format is accepted — **PDF, Word, HTML, or any text-based format (Markdown, etc.)** — committed in the repository (e.g. report/).

2. The presentation supports

The slides (and any other material) you use for the final demo: **10 min presentation + 10 min questions**.

Format: **any format is fine** (PDF, Word, the original slides file, etc.), committed in the repository (e.g. slides/).

3. The code + documentation on how to run it

- **All your source code:** machine discovery, deployment, cleaning, servers, client, map/reduce, fault tolerance, Kafka scripts, etc.
- A **README** that explains how to run everything **from scratch**:
 - prerequisites (languages, versions, environment),
 - how to generate the list of machines,
 - how to **deploy**, how to **run a job**, how to **clean**,
 - where the inputs come from and where the outputs land.
- Anyone (including me) must be able to **follow the README and reproduce a run**.
- **Do not commit** large datasets (Common Crawl splits) or build artifacts — add a `.gitignore`.

4. The completed self-assessment questionnaire

Take the **self-assessment criteria** document, fill it in **honestly** (tick the boxes, answer the wrap-up questions), and commit it.

Format: the filled-in **Markdown / PDF / Word**, committed in the repository (e.g. `self-assessment.md`).

Suggested repository layout

```
your-repo/
├── README.md           # how to deploy / run / clean (start here)
├── .gitignore          # exclude datasets, outputs, build artifacts
├── src/               # all your source code
│   ├── deploy/        # deployment + cleaning scripts
│   ├── server/        # server code
│   ├── client/        # client code
│   ├── mapreduce/     # map / shuffle / reduce
│   └── kafka/         # Kafka Streams scripts (deploy / clean / word
count)
├── report/
│   └── report.pdf      # PDF, Word, HTML, Markdown... any readable form
at
├── slides/
│   └── presentation.pdf # any format is fine
└── self-assessment.md  # completed questionnaire
```

Submission checklist

- ☐ **Report** committed (any readable format), covering the 7 required points
- ☐ **Presentation slides** committed (any format)

- ☐ **Code** committed with a **working README** (deploy / run / clean)
- ☐ **Completed self-assessment** questionnaire committed
- ☐ **No** large datasets / build artifacts committed (.gitignore in place)
- ☐ I **have access** to the repository (or the URL was sent to me)
- ☐ Everything **pushed at the latest one week after the last session**